STAT 312. Basic Statistics for Engineering and Science

Case Western Reserve University, Fall 2026

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- **Section 103**: M/W/F 10:25-11:15am, White Building 411
- **Section 104**: M/W/F 11:30-12:20pm, White Building 411
- **Section 105**: M/W/F 02:05-02:55pm, Nord Hall 410

Welcome to STAT 312! Engineers and scientists routinely work with data that contain variability and uncertainty. For example, how can we determine whether a change in a manufacturing process actually improves product quality? How can we use measurements from a sample to draw conclusions about a larger population? How can we quantify uncertainty when comparing two designs, estimating an unknown quantity, or predicting future performance?

Statistics and probability provide a framework for answering these questions. They allow us to collect and summarize data, understand variability, quantify uncertainty, draw meaningful conclusions, and make informed decisions.

Throughout the semester, we will develop the fundamental statistical and probabilistic tools used in **engineering and scientific problem solving**. Topics will include descriptive statistics, probability and probability distributions, sampling distributions, estimation, confidence intervals, hypothesis testing, and regression analysis. Our emphasis will be on developing statistical reasoning and intuition, understanding when and why particular methods are appropriate, and applying those methods to real-world problems. Rather than simply memorizing formulas, you will learn to select appropriate statistical methods, carry out analyses, interpret results, evaluate assumptions, and communicate conclusions in the context of the problem being studied.

Learning Objectives

Upon completion of this course, students should be able to:

- Build, use, and interpret **graphical and numerical summaries** of data.
- Apply fundamental concepts of **probability**, including conditional probability, independence, random variables, and probability distributions, to engineering and scientific problems.
- Identify and work with commonly used **discrete and continuous probability distributions**, and recognize situations in which they provide appropriate models.
- Understand **sampling variability and sampling distributions**, including the role of the Central Limit Theorem in statistical inference.
- Use sample data to **estimate unknown population parameters** and quantify uncertainty using standard errors and confidence intervals.
- Formulate appropriate **statistical hypotheses**, conduct hypothesis tests, interpret p-values, and draw conclusions in the context of an engineering or scientific problem.
- Build, interpret, and evaluate **simple linear regression and correlation models**, and use regression models for estimation and prediction.
- Use appropriate **statistical tools and software**, when needed, to support statistical calculations, data analysis, and interpretation.
- **Interpret and communicate statistical results** clearly using appropriate terminology, notation, reasoning, and evidence.

Instructor/Grader Team

About your professor

Md Mutasim Billah, PhD

Pronunciation: listen here

Office: Sears 569

Email: mdmutasim.billah@case.edu

Notes from "your instructor"

Greetings! You can call me *Bill* or *Dr. Billah* or *Professor Billah* & I use he/him pronouns. Back when I was an undergrad student, I didn't have the best experience in Intro Stat - those courses often emphasized formulas over real understanding. That experience has shaped my teaching - I concentrate on illustrating how statistical theories connect and can be applied in the real world. I'm excited to teach STAT 312 and to create a more meaningful experience - one that helps all students feel confident applying it beyond the classroom. My methodological research lies at the intersection of statistical genetics, biostatistics, and genomics. My current research interests include developing novel statistical methods and computationally efficient bioinformatics tools, leveraging modern machine- and deep-learning approaches analyze high-dimensional next-generation sequencing and multi-omics data to identify genes and regulatory mechanisms underlying complex diseases. Outside of my academic work, I enjoy spending time outdoors with family and friends or cooking variety of foods. If you can't find me anywhere, I might be busy playing soccer or exploring new worlds on my PS5 Pro!

Office hours:

- **Location:** My office (Sears 569)
- **Times:** M/W: 09:00 - 10:00 am and 12:45 - 1:45pm
- **Email Response Time:** I do my best to reply to emails promptly during weekdays. Please note that messages sent after 4:00 pm or on weekends may take longer to receive a response.

About your Graders

We have three wonderful **STAT 312 graders** this semester! Their primary role is to assist with the **assessment and grading of course assignments** and, when appropriate, portions of exams and other course assessments.

The graders will follow grading guidelines established by the instructor to help ensure that student work is evaluated consistently and fairly. Please keep in mind that graders do not determine course policies, assignment requirements, grading schemes, or final course grades.

For questions related to grading, please contact the grader who graded that specific problem. For example, if a grader was responsible for Problem 1, please contact that grader only for questions about Problem 1. If your question involves a course policy, extension, accommodation, academic integrity concern, or another matter requiring an instructor-level decision, please contact me directly.

In selecting and working with our graders, we emphasize **fairness, professionalism, kindness, and respect**. I expect the same from students in all interactions with them. Please

respect the time, effort, and important contribution of our graders in supporting STAT 312 throughout the semester.

Course resources

Textbooks & Online Course Manual

- Applied Statistics and Probability for Engineers, 7th edition, by Montgomery and Runger. You can download an e-Book from Canvas (Files).
- Statistical Methods for Engineers, by Vining and Kowalski. You can download an e-Book from Canvas (Files).
- The online course website includes all the course resources.
- See the rough schedule of the course, which will be updated throughout the semester.

Canvas

Canvas includes announcements, general resources, submission links, feedback, and a discussion forum for student questions.

Statistical software

We will use the (completely free and open source) R programming language for some data analysis in this course. RStudio (an interface for R) facilitates the use of R.

- Desktop version: Download for Windows or Mac at https://posit.co/downloads/. Note: You first need to download and install R on your computer in order to use the desktop version of RStudio
- Library: Resources offered by Kelvin Smith Library are available here

More detailed instructions on downloading, installing, and getting started with R, and RStudio are available on the R Resources tab.

Communication & Thriving

Asking questions/communicating

Office Hours

OH are a great place to chat about the course, career planning, life,… Please visit me!!

- OH are oriented around group discussion. They are not first come, first served appointments.
- Since it's not an effective way to deepen your learning, OH are not a place to sit and do assignments with me. It's an opportunity to discuss concepts & specific questions.

Canvas Discussion: STAT 312 Discussion Forum

This forum is where you'll communicate outside class. Students can post and answer comments / questions there. This is an informal way to converse, ask questions, share info, & connect. Do not rely on receiving responses outside weekdays between 8am & 4pm.

STAT 312 Tutor - Walk-in

- Offered by the Department of Mathematics, Applied Mathematics, and Statistics
- **Fall 2026 Tutor:** Yiming Chen (Sears 543)
  - **Hours:** Mon (9:30–11:00 AM), Tue (4:00–5:00 PM), Wed (9:30–11:00 AM), Thu (4:00–5:00 PM)

Peer Tutors - One-on-one

- Offered by the School
- Students have free access to peer tutors for STAT 312.
- Appointments can be scheduled through tutortrac. Click here for more instruction

What to do when you have a question for me?

- If it's non-private (e.g. about policies, HW assignments, practice problems, etc), you should post it on STAT 312 Discussion Forum on Canvas. Remember- collaboration is the KEY!
- If it's personal (e.g. about an absence, concept understanding, etc), email me.

- It's good, professional practice to check whether your question is already answered in the provided resources. For example:
  - **Info** (what to do if you miss class): syllabus
  - **Due dates**: course schedule in the online manual + announcement email
  - **Exam dates**: course schedule in the online manual + announcement email
  - **Homework & Quiz policies**: STAT 312 HW Assignment & Quiz policies

Thriving in STAT 312

Plan Ahead

You should plan to spend ~6-9 hours on any 3-credit course, including class time. Stay up-to-date on the course schedule and carve out time for studying, doing homework, & participate in quizzes.

Do the Things

At minimum, thriving in this course requires the completion of some concrete tasks. Complete all assignments, participate in the quizzes, regularly attend & engage in class, complete practice problems (PP), and check the PP solutions.

Build a Foundation

If your main focus is on checking off some boxes, you won't get much out of this course (or college in general). Deeper, enduring learning requires more. Carve out time to rewrite, reflect upon, & review your notes. Summarize concepts in your own words.

Engage, Ask Questions, Have Fun

Actively participate in the class & take ownership of your learning. PLEASE: Don't be afraid to ask for help, make mistakes, and ask questions! These skills are critical to your well-being & learning. Finally, have some fun, be curious, and reflect upon what surprises you about the material and yourself

Course Policies

Flexibility

- **Missed Class**: Regular attendance & active participation are strongly encouraged and expected. In my experience, students who attend STAT classes consistently tend to finish the course with a deeper understanding of the material, greater satisfaction with their learning, and stronger academic performance.
- **Homework Assignments & Quiz**: Please *carefully* go through all the sections from STAT 312 HW Assignment & Quiz policies doc.
- **Exams**: No revisions. Additional flexibility will be provided in rare extenuating circumstances, upon discussion. Exceptions must be discussed with me (not assumed) early on (not after the fact).

PLEASE REACH OUT WHEN YOU NEED HELP.

Absences

What to Do If You Miss Class

1. **Send me a quick email.** You do not need to share a reason for your absence, especially if it's personal. It's just a simple courtesy & keeps communication lines open.
2. **Check the Course Schedule** in the course website for what is happening in class that day.
3. **Go through the lecture notes (from course website activities), class notes, and solved handouts on your own** (I usually post the class notes and handouts by the evening of the class day). Also, be sure to review the related topics in the **textbooks**, as some details discussed in class may not appear in the class notes. For example, I may work through additional examples on the board or explain certain concepts verbally. **You are responsible for these materials as well, and usually they appear on quizzes, HW Assignments, or Exams.**
4. After finishing all the steps mentioned above, **ask any follow-up questions** on the Canvas Discussion forum or come to office hours (OH).

Artificial Intelligence (AI)

Read the Use of AI sub-section from STAT 312 HW Assignment & Quiz policies doc.

Community & Academic Integrity

Read the Academic integrity policies from STAT 312 HW Assignment & Quiz policies.

The 5 Core Components

(1) Engagement

Engagement is important to your own learning & to fostering a supportive learning community.

! Expectations

- **Do not miss in-person class sessions (except in emergency situations).**
- **When attending class:**
  - Be on time & don't leave early
  - Do not use your phone (phones must be put away when you enter the classroom, even if class hasn't started)
  - Do not use your laptop for anything other than taking notes and in-class activities (e.g., no videos, no email, no messaging apps, etc.).
  - If you plan to use a laptop for note-taking, please obtain my permission first. Otherwise, I may ask you to put your laptop away at any point during class if its use appears unrelated to the course.
  - Be actively present (e.g., don't work on other courses, etc.)
- **Course handouts:**
  - Handouts are available for most units in this course and will be posted throughout the course in the course website.
  - You can download the handouts from the **Activities** tab by going to the corresponding unit in the Course Website.
  - These handouts include examples that we will work through together in class after discussing and writing down the related concepts, rules, and theories.
  - If you use **paper and pencil**, you are **REQUIRED to print the handout and bring it to class**.

  - If you use a **tablet, iPad, or other writing device**, you may download the handout directly to your device.
  - I strongly recommend creating a folder named **STAT 312** on your device and organizing the handouts into separate folders for each unit (e.g., `Unit 01`, `Unit 02`, etc.).

- **Class activities**
  - On selected class days, we will complete short in-class activities. Students who are present and participate in these activities will receive activity points, and those points will be added to as a separate quiz score (for example, Quiz A).
  - If you are unable to attend a class, please send me an email before class begins with your section number (see the syllabus for the section number, if you are unsure). If that happens to be a day when we complete a Class Activity, you may otherwise miss the opportunity to earn those activity points.
  - Please note that requests regarding missed Class Activity points cannot be processed unless your section number is included in the email. Requests submitted after the class has ended will also not be processed.

- **Outside class:**
  - Check your email for Canvas announcements and stay updated on the Canvas Discussion Forum.
  - When you have questions, or just want to chat, please stop by OH!

(2) Practice Problems (PP)

Complete the **Practice Problems (PP)** before attempting the HW assignments. Working through the PP alongside the lecture notes and class materials will deepen your understanding of the core concepts and help you feel more confident when tackling homework problems. Consistent practice will also make your exam preparation easier and more effective.

There are two types of Practice Problems:

Required Practice Problems

Complete the **Required Practice Problems** throughout the unit as we cover the corresponding topics in class.

These problems reinforce the concepts discussed in the lecture notes, handouts, and other course resources.

Before attempting the HW assignments and preparing for exams, make sure you have completed and reviewed all Required Practice Problems from the unit.

Supplemental Practice Problems

The **Supplemental Practice Problems** provide additional practice opportunities for students who would like more practice with the concepts covered in the unit.

Complete the Required Practice Problems first, and then use the Supplemental Practice Problems as needed for additional review and practice.

(3) Quizzes

Each quiz assesses your general understanding of the topics covered during the corresponding unit. Detailed directions and policies are here: STAT 312 HW Assignment & Quiz policies.

(3) HW Assignments

In the HW assignments, you will practice and explore the course material in more depth. Detailed directions and policies are here: STAT 312 HW Assignment & Quiz policies.

(4) Exams (Content Expertise)

Your course engagement, collaboration, preparation, practice, and application will support your deeper understanding of the course material. This will be assessed through three in-person exams. You must schedule all travel and other commitments around them - there will not be any alternative exam times.

- **Midterm Exam 1:** Monday, Oct 5
- **Midterm Exam 2:** Monday, Nov 16
- **Final Exam (Comprehensive):** *Finals week*
  - The final exam will be written to take ~1.25 hours, but you will have the full 2-hour period to complete it.
    - 10:25–11:15am section 103: Dec 14, 8:00-10:00am in class
    - 11:30am–12:20pm section 104: Dec 9, 8:00-10:00am in class
    - 2:05–2:55pm section 105: Dec 16, 12:00-2:00pm in class

Exam Policies

- **All Exams will have the following format:**
  - Taken individually, using pen/pencil & paper
  - You may use a physical calculator (not the calculator in your phone) in the exam
  - Closed notes, but you may use one sheet of paper, with writing on one side (two sided writing is NOT permitted) for the midterm exams. These can be handwritten or typed, but you cannot include screenshots or share formula sheet. Making your own Formula Sheet is important to the review process - as you are required to submit the Formula Sheet along with the answer paper.
- **Final Exam will be comprehensive.** For the final exam, you may use one sheet of paper, with writing in both sides.

Calculating Final Grades

Course Percentage

20% HW Assignments +
10% Quizzes +
40% Midterm Exams (20% each) +
30% Final Exam (Comprehensive)

Final Grade Scale

Grade	Course percentage
A	90–100%
B	80–89.99%
C	70–79.99%
D	60–69.99%
F	59.99% and below

Other policies

Religious Observance

Students may wish to take part in religious observances that occur during the semester. If you have a religious observance/practice that conflicts with your participation in the course, please contact me before the end of the second week of the semester to discuss appropriate accommodations.

In an effort to respect religious diversity, I request that students who plan to observe a religious holiday during scheduled class meetings/class requirements talk to me about reasonable consideration by the end of the second week of the course.

Well-being

I want you to succeed. Both here at CWRU and beyond. To help make this happen, I am committed to the following.

Respect: Everyone comes from a different path through life, and it is our moral duty as human beings to listen to each other without judgment and to respect one another. I have no tolerance for discrimination of any kind, in and out of the classroom. If you are seeking campus resources regarding discrimination, the Office of Equity and the Division of Student Affairs are wonderful resources.

Accommodations: If you need accommodations for any reason, please contact Disability Resources to discuss your needs, and speak with me as soon as possible afterwards so that we can discuss your accommodation plan. If you already have official accommodations, please discuss these with me within the first week of class so that you get off to a great start. Contact me if you have other special circumstances. I will find resources for you.

Title IX: You deserve a community free from discrimination, sexual harassment, hostility, sexual assault, domestic violence, dating violence, and stalking. If you or anyone you know has experienced harassment or discrimination, know that you are not alone. CWRU provides staff and resources to help you find support. **Please be aware that as a faculty member, it is my responsibility to report disclosure about sexual harassment, sexual misconduct, relationship violence, and stalking to the Title IX Office.** The purpose of this report is to ensure that anyone experiencing harm receives the resources and support they need. I will keep this information private, and it will not be shared beyond this required report.

You may also contact CWRU's Title IX Coordinator directly (phone: 216-368-3066; e-mail: rachel.lutner@case.edu); they will provide you with supportive measures, resources, and referrals. Additional information about how to file a report (including anonymously) is available on the Title IX website.

General Health and Well-being: I care that you prioritize your well-being in this semester and beyond. Investing time into taking care of yourself will have profound impacts on all aspects of your life. Remember that beyond being a student, you are a human being carrying your own experiences, thoughts, emotions, and identities. It is important to acknowledge any stressors you may be facing, which can be mental, emotional, physical, cultural, financial, etc., and how they can have an impact on you. I encourage you to remember that you have a body with needs. In the classroom, eat when you are hungry, drink water, use the restroom, and step out if you are upset and need some air. Please do what is necessary so long as it does not impede your or others' ability to be mentally and emotionally present in the course. Outside of the classroom, sleeping well, moving your body, and connecting with others can be strategies can help nourish you. If you are having difficulties maintaining your well-being, please don't hesitate to contact me and/or find support from physical and mental health resources hereand here.

Want 'A' in this course? Read!

Ask Questions

When you have questions, please stop me during class, ask your neighbor, post question to the Discussion Forum on canvas, email me and come to office hours. Saying *"I don't understand"* is an important part of learning and it helps others around you.

Come to Office Hours

Office hours are a great time to talk about course material, study strategies, selecting courses, declaring a major, grad school and/or career planning, or life in general. You don't need to have a specific question in order to attend: it can also be a space to review concepts, talk through examples, or just chat!

Make Time

Performing a thoughtful statistical analysis requires time: to plan, to implement, to interpret, and to revise.

- Start your assignments early. It is very hard to be creative when you are in a rush.
- In addition to the roughly 3 hours we spend together during class, expect to spend about **6-9 hours per week** on this class.
- Don't miss any practice problems - they are designed to give you a core understanding of the subject matter!

Attend Class

Active participation in this class will be key to your learning. To ensure the best learning experience for you and your classmates:
- Come prepared, review the previous class resources as required - Engage in class by taking notes, group works, asking and answering questions - Make full use of the entire class period
If you can't attend:
- Check Canvas and Course Website to see what you missed
- Review the lecture and class materials - Complete the practice problems - (After doing all of the above) come to office hours with specific questions.

Prioritize Your Well-Being

Investing time into taking care of yourself will help you engage more fully in your academic experience. Remember: beyond being a student, you are a human being carrying your own experiences, thoughts, emotions, and identities with you. If you are having difficulties maintaining your well-being, please contact me and/or check out available campus resources (mentioned in the 'Other Policies' part of the syllabus. As part of prioritizing your well-being (and others'), **please stay home if you are feeling sick.** Follow the recommendations above (*Attend Class*) and below (*Communicate*) if you miss class.

Communicate

I will do my best to clearly communicate changes to expectations, deadlines, office hours, or class meetings due to instructor illness or unexpected life issues.
Please check Canvas announcement regularly so you don't miss announcements.
I also ask that you check in with me as soon as possible if:
- You need to miss multiple classes in a row
- You have a conflict (e.g., athletic competition, religious observance) with a scheduled exam - You need accommodation(s)
- Something about the class is not working for you

Tips from Past Students!

Here are some comments from previous students in my **Engineering Statistics** class that may help you succeed in STAT 312 as well!

- Pay attention to the class lectures, and take notes. Makes the entire class 10x easier.
- Go to the office hours. He is a really kind professor, and will help so much during

the course…getting to know professor Billah is just a joy and can make the course so much more fun, not to mention he will definitely make your day.

- Class is easy to follow along, content is easy to follow, homework and quiz help prepare you for the exam.
- Discussion forum on canvas will help you to secure maximum points, don't hesitate to ask questions there; also if you know the answer, let others know as well!
- I liked the way the class was setup as a whole. The discussions were helpful.
- Discussions were helpful to see that you had a similar understanding as your classmates.
- My advice would be to attend office hours! Dr Bill gives great feedback. The meetings are short too, so no need to worry about allocating a significant amount of time.
- Personally, this was my favorite class of the semester. More as how accessible the assignments and lectures were. I could re-watch lectures and with the required discussion forum, I was able to get help by others in the class if I didn't get the right answer, or faced a problem.
- The announcement emails to keep students on track as well as other call outs were nice throughout the semester. I think overall, work and requirements were well done and distributed.
- Continue with the discussion posts because they made problems easier to solve and when you get the same answer as another student in the class, you feel more confident in yourself.
- I really liked the handouts and the notes that you had us take to better understand the material. Hearing different things from other classes, it seemed like this class was the most prepared and I knew and could explain more topics better than other students in other classes. The live lecture videos of the class were great too to go back and review certain things if I was confused. Overall the class was great!
- Thank you for teaching the course! I like how organized your lecture notes were. They were easy to read and allowed me to easily copy them down for my own reference. You were also very great to talk with and were enthusiastic about the course. Please keep that incredible attitude, as students will appreciate that very much!
- I like how you tell us what the material is with very detailed information. Then, we jump to exercises to further understand the concept. I personally have found that you bring a lot of life into the subject matter by giving funny but relevant examples and problems to solve in class. I enjoyed the way that you taught the subject matter and you were knowledgeable enough to answer questions that students had. I recommend keeping playing music at the beginning of class especially because I personally feel more interested in the class because of it and I think others do too.
- I think my experience in this class was so good because of the high level of engagement with students. The class was made fun and the material was applicable.

> Notes were very neat and well organized, which also helped to make learning the material a lot easier. The pace of the course was perfect too in order to really learn the first time without always having to look back at the Huskycast. Again, I liked the personal interaction with the class a lot, it made me want to come to class more than out of pure obligation. Thank you for a great semester!

Note

This syllabus is subject to change at any time! Announcements of changes will be made in class or via announcement on Canvas.